Money Multiplier Theory

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1 Money

Money, largely originated as a response to the limitations of the bartering system [Middelkoop, 2014]. To counter these shortcomings, money serves several essential functions: it acts as a **medium of exchange**, a **measure of value** and a **store of value**. [Gronow, 2020]

A contingent or derived function of money is that it serves as the **basis of credit** [Suman, 2015]. Money being the store of value, enables one to **lend** or **borrow**. **Credit** is an agreement between a lender and a borrower that allows the borrower to obtain funds, goods or services now and repay them later. But is this agreement always robust? If a person lends money to a borrower, is it always guaranteed that they will be paid out?

2 Banks as Financial Intermediaries

The answer is undoubtedly NO. **Credit Risk** arises from the potential that a borrower will not repay the respective lender. The practice of lending and borrowing between only trusted individuals would restrict money's function as the basis of credit. Banks acting as **financial intermediaries** work to reduce this risk.

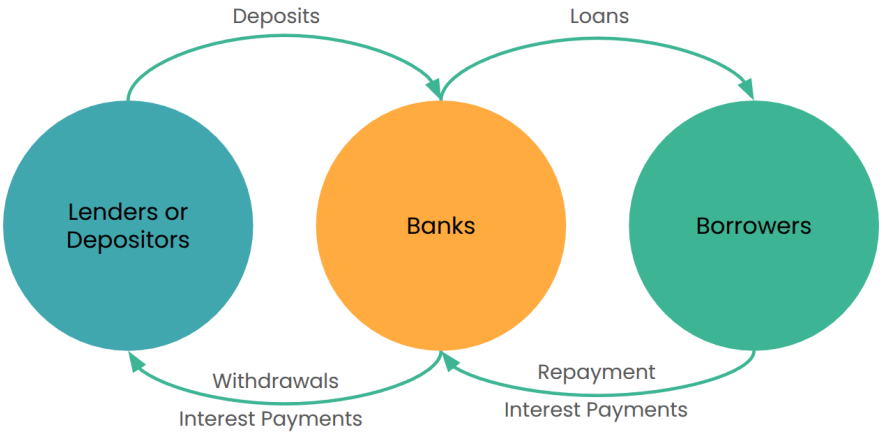


Figure 1: Bank's role as an intermediary

Apart from this, banks also function as a trusted place for **depositing money**. These deposits (**debit deposits**) can then be utilized for lending purposes, under the assumption that not all depositors will withdraw their funds at the same time. As banks earn interest from loans, they would like to maximize the amount of money they are lending. At the same time, the ability to repay the depositors on demand is also crucial for a bank's survival. Hence, a bank must balance its lending activities while ensuring that sufficient funds (**reserves**) remain to repay depositors [Williams, 2010]. **Central Bank** is an important institution for a country's economy which along with being a (**commercial**) bank as defined above, also performs important activities like issuing currency, regulating reserve ratios, and so on.

The relationship between central banks and commercial banks serves as a foundation for **credit creation**.

3 The Theory

3.1 *Definitions*

- **Deposits**: Money stored by an individual at a bank for future financial transactions.
- **Reserves**: Reserves is defined as the amount of money a bank can't use for lending purposes.
- **Reserve Ratio** (rr): The percentage of value of deposits kept in reserves by a bank.
- **Monetary Base**: $M_0 = C + R$. C is the total Currency in Circulation and R is total amount of reserves with the central bank. It is also referred to as *high-powered money*.
- **Money Supply**: $M_1 = C + D$. D is the total value of deposits present in banks.

[Dornbusch et al., 2004]

3.2 *How is credit created?*

Suppose Amar deposits **Rs. 1000** (D_1) in Bank B. For simplicity, assume this is an imaginary economy where this **Rs. 1000** actually increases the monetary base (does not belong to the currency in circulation), so $\Delta M_0 = D_1$. This gives the bank deposits worth **Rs. 1000**. With a reserve ratio (rr) of 10%, Bank B must keep **Rs. 100** in reserve and can lend out the remaining **Rs. 900**.
Next, Akbar borrows **Rs. 900** from Bank B, which is credited to his account. Now, total deposits in Bank B are **Rs. 1900**, **Rs. 1000** in Amar's account and **Rs. 900** in Akbar's account. Accordingly, required reserves are **Rs. 190** (10% of **Rs. 1900**). Since the bank has already lent out **Rs. 900**, it can lend an additional **Rs. 810**.
Anthony then borrows this **Rs. 810**, and it is credited to his account. Deposits now total **Rs. 2710** (**Rs. 1000** + **Rs. 900** + **Rs. 810**). Reserves required are **Rs. 271**. From this, Bank B can lend a further **Rs. 729**.
This cycle will continue. D increases with every round of lending, this directly increases the money supply, M_1.

Round	Total Deposits (Rs.)	Required Reserves (Rs.)	Excess Lending (Rs.)
1	**1000**	**100**	**900**
2	**1900**	**190**	**810**
3	**2710**	**271**	**729**
4	**3439**	**343.9**	**656.1**
.	.	.	.
.	.	.	.
Last	**????**	**1000**	**0**

This process of lending continues until the total required reserves match the initial deposit of **Rs. 1,000**. The total increase in deposits (ΔM_1) can be calculated as the sum of an infinite geometric series:

$$\Delta M_1 = D_1 + (1 - rr) \times D_1 + (1 - rr)^2 \times D_1 + \dots$$

Where D_1 is the initial deposit (Rs. 1,000) and rr is the reserve ratio (0.1). This series simplifies to:

$$\Delta M_1 = \frac{D_1}{rr} = \frac{1000}{0.1} = \textbf{Rs. 10,000}$$

So, when the initial money (M_0) increases by **Rs. 1,000**, the total money supply (M_1) increases by **Rs. 10,000**.

The ratio of the change in the total money supply to the initial change in reserves is known as the **money multiplier (γ)**.

$$\gamma = \frac{\Delta M_1}{\Delta M_0} = \frac{D_1/rr}{D_1} = \frac{1}{rr}$$

In our example, the multiplier is **10** (1/0.1), this implies that a **Rs. 1** increase in the monetary base (M_0) leads to **Rs. 10** increase in the money supply (M_1). Here, we assumed that the currency in circulation did not get affected by Amar's deposit. We extend the model to accomodate the currency in circulation. [Tabarrok, 2017]

3.3 *Towards Reality*

We define the money multiplier (γ) the same as above,

$$\gamma = \frac{\Delta M_1}{\Delta M_0}$$

But now, $\Delta M_1 = \Delta C + \Delta D$, and $\Delta M_0 = \Delta C + \Delta R$. By dividing the numerator and denominator with ΔD, γ simplifies to

$$\gamma = \frac{\frac{\Delta C}{\Delta D} + \frac{\Delta D}{\Delta D}}{\frac{\Delta C}{\Delta D} + \frac{\Delta R}{\Delta D}}$$

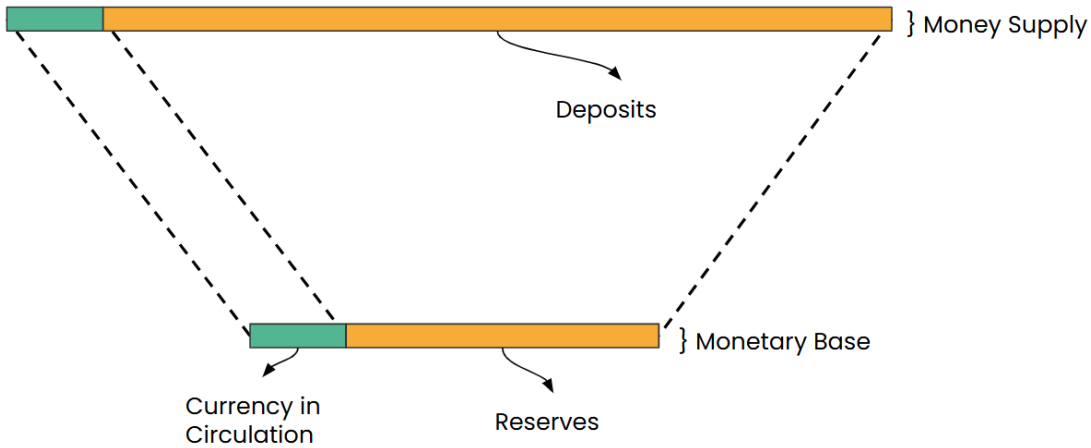


Figure 2: How money is multiplied

So,

$$\gamma = \frac{cr + 1}{cr + rr}$$

where cr is the public's preference for holding currency (formally called currency-deposit ratio) [Dornbusch et al., 2004]. It is now evident that the central bank, by having the power to manipulate the reserve ratio, is an important determinant of the money multiplier and, consequently, the nation's money supply.

3.4 *The role of Central Banks*

The Reserve Bank of India mandates commercial banks to have two types of reserves:

- **Cash Reserve Ratio (CRR)**: This refers to the proportion of total deposits of the commercial bank which they must keep as cash reserves with the RBI.
- **Statutory Liquidity Ratio (SLR)**: It is the ratio of liquid assets which all commercial banks have to keep in the form of cash, gold and unencumbered approved securities.

The RBI has control over these values through monetary policy. While both CRR and SLR are key monetary policy instruments, the value of CRR is often considered to be the value of rr in the money multiplier process. This is because CRR funds are "locked" deposits kept with the Reserve Bank of India (RBI), making them completely unavailable for lending. In contrast, SLR is maintained through interest-earning government securities, meaning these funds are not fully withdrawn from the financial system.

Both ratios are vital for managing liquidity in the economy. To combat inflation, the RBI increases the CRR and SLR to reduce the money supply available for lending. Conversely, to counter deflation or stimulate growth, it lowers these ratios. The SLR serves an additional

purpose: by compelling banks to invest in government securities, it provides a stable source of funding for the government's fiscal requirements [Rani, 2019]. On June 6th, 2025 the RBI slashed the CRR from 4% to 3% causing a **Rs. 2.5 Lakh Crore** rise in lending capacity of commercial banks. The SLR remained constant at 18% [PTI, 2025].

3.5 *Criticisms*

The 2008 financial crisis served as an illustration of the shortcomings of the money multiplier theory. As a response to the crisis, the Fed (central bank of U.S) significantly expanded the monetary base, which did not get reflected in lending (deposits) [Abel et al., 2016].

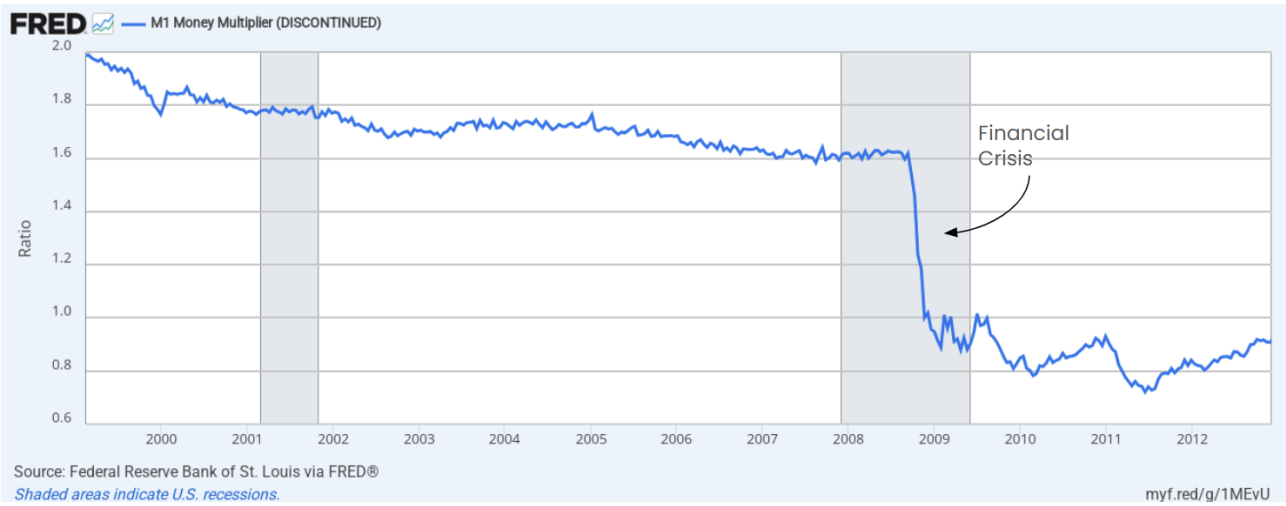


Figure 3: Trend of Money Multiplier Ratio ($\frac{M_1}{M_0}$)

This raised doubts on the ability of central banks to control the money supply. The **endogenous money theory** contends that creation of money is driven by **credit demand**, with loans creating deposits rather than the other way around. Because the money-multiplier model presumes that deposits come first and then get multiplied through lending, this claim undermines its very foundation. Banks **create credit** when they make loans, they don't lend out deposits, they don't lend out reserves to customers in the form of loans, nor do they have to wait for deposits to come in in the first place. Banks **can** fund loans with deposits, but irrespective of the deposit funds available banks approve loans to creditworthy borrowers with a stroke of a button (thanks to electronic money) [Markert, 2021]. Another criticism placed is the difficulty and ambiguity in measuring the money supply. The RBI itself defines four different types of money supplies[Abel et al., 2016].

4 Concluding Summary

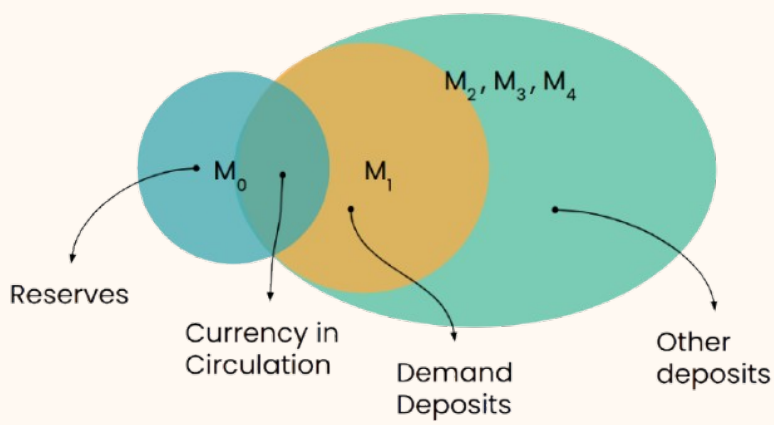
Money began as a way to get around the limits of bartering, but it also serves as the basis credit. Banks take deposits, keep a slice as reserves and lend the rest, which in theory multiplies through the system — this is the result of the money-multiplier theory. The central bank

influences how big that effect can be by changing reserve requirements and liquidity rules like the CRR and SLR.

In practice, though, lending doesn't always line up neatly with reserves. After events like the 2008 crisis, banks held more reserves but didn't automatically make more loans, showing that credit demand and risk appetite matter too. Still, the multiplier remains a useful framework for understanding how policy changes filter through the banking system; it just doesn't capture the whole picture.

Money Supply

It is the total amount of money in circulation or in existence in a country. The Reserve Bank of India (Central Bank of India) proposes four alternative measures of money supply:



- M_0 : Monetary Base: Reserves in Banks + Currency in Circulation
- M_1 : M_0 + Demand Deposits (Deposits you can withdraw at any time)
- M_2, M_3, M_4: M_1 + other types of deposits

As we move from M_0 to M_4, the **liquidity** (how easily it could be converted to cash without disrupting the value of money) of the money supplies decrease.

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